

Implicit Neural Representations for 3D LSFM Distortion Correction via Axial Vision Transformers

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1 Introduction

Light-Sheet Fluorescence Microscopy (LSFM) has become an indispensable tool for rapid, high-resolution volumetric imaging of biological samples. However, LSFM inherently suffers from sample-induced optical aberrations and scattering, which severely degrade the axial resolution and produce elongation artifacts.

Recent computational imaging approaches have sought to address scattering and optical aberrations by jointly estimating the object and the unknown distortion fields. For instance, Xue et al. (2022) [1] proposed BRIEF, an epi-mode microscopy technique that reconstructs both 3D fluorescence and 3D refractive index (RI) distributions from a single dataset to digitally correct multiple-scattering effects. Similarly, Feng et al. (2023) [2] developed NeuWS (Neural Wavefront Shaping), a guidestar-free method that leverages continuous neural signal representations and maximum likelihood estimation to image through static and dynamic scattering media.

While these methods demonstrate the power of joint optimization and implicit neural representations for aberration correction, they predominantly rely on either explicit multiple-scattering forward models requiring specialized acquisition setups [1] or coordinate-based MLPs optimized from scratch for each specific scene and aberration instance [2], limiting their generalizability and rapid adaptation to new datasets.

In contrast, my proposed framework introduces a novel, amortized approach for 3D LSFM distortion correction by synergistically combining a Vision Transformer (ViT) encoder with a Multi-Head Implicit Neural Representation (INR):

- **Global Context via ViT Latent Space:** Unlike scene-specific optimization methods, my ViT encoder processes the distorted stack as an axial sequence, capturing the global context of the z -axis elongation artifact. By mapping the volume into a compact global latent space, the model learns the underlying distribution of distortions. This enables powerful *transfer learning*; the amortized network can be pre-trained on diverse synthetic calibration data and then rapidly fine-tuned on real, sparse experimental data, avoiding the need for exhaustive per-volume optimization.
- **Multi-Head Disentangled INR:** Building upon the continuous representation capabilities seen in methods like NeuWS, my Multi-Head INR goes a step further by explicitly disentangling the reconstruction tasks. Guided by the global latent vector, the network separates the prediction of the corrected fluorescence density from the estimation of the geometric distortion subspace (via an auxiliary branch). Coupled with local coordinate evidence sampling, this structured conditioning effectively resolves the ill-posed inverse problem, ensuring stable and high-fidelity restoration of sparse biological structures.

2 Synthetic 3D LSFM Bead Calibration Dataset

To train the restoration models, a reproducible synthetic forward model is utilized to generate paired datasets of distorted observations x and corrected targets y . This pipeline simulates the physical degradation processes of LSFM and provides identical supervised signals to real bead calibration protocols. My synthetic dataset generation approach is fundamentally inspired by the calibration logic proposed by Julia et al. [3], adapting their real-world bead segmentation and axial elongation protocol into a fully reproducible synthetic forward model.

2.1 Bead Scene Sampling and Corrected Target

For each sample, beads are randomly distributed within the Field of View. The signed distance for a spherical bead is defined as $d_k(p) = \sqrt{r_k^{-2}((z - c_z)^2 + (y - c_y)^2 + (x - c_x)^2)} - 1$. To model a smooth bead boundary, a sigmoid function is applied: $b_k(p) = \sigma(-\alpha d_k(p))$ with $\alpha = 8$. The raw corrected volume $\tilde{y}(p) = \sum_{k=1}^K a_k b_k(p)$ is then min-max normalized to $[0, 1]$ to yield the final target $y(p)$.

2.2 Axial Distortion and Asymmetric RI Mismatch

The primary LSFM distortion is modeled as a z -axis elongation. A global stack-level distortion factor f_{base} is sampled, and individual bead distortions are assigned via f_k . The elongated bead assumes a prolate ellipsoid shape with axial radius $r_{z,k} = r_k \cdot f_k$. To emulate complex refractive index mismatches in cleared samples, an optional spatially varying distortion field is applied using a low-order polynomial field over normalized coordinates.

2.3 LSFM Degradation Operator

The distorted volume $\tilde{v}(p)$ undergoes successive degradations to simulate the microscope’s forward path, including Light-sheet Illumination (Gaussian envelope), 3D PSF Blur (separable anisotropic 3D Gaussian kernel), and Background/Noise (mixed Poisson-Gaussian distribution). The complete forward model is summarized as:

$$x = N \left(\text{PoissonGaus} \left(G * (L \cdot v(f, c^{\text{obs}})) + b \right) \right) \quad (1)$$

3 Model Architecture

Figure 1 illustrates the overall architecture of the proposed amortized ViT-INR framework. The network conceptually integrates a global context pathway for understanding macroscopic distortion via a Vision Transformer and a continuous local reconstruction pathway via a Multi-Head Implicit Neural Representation.

3.1 Vision Transformer (ViT) Encoder and Latent Space

The distorted stack is formulated as a sequence of axial slices along the z -axis:

$$x \in \mathbb{R}^{1 \times Z \times Y \times X} \implies S = [S_1, S_2, \dots, S_Z] \quad (2)$$

The input sequence is projected via a linear embedding layer and augmented with a learnable class token and positional embeddings. The sequence is processed through L consecutive Multi-Head Self-Attention blocks:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \quad (3)$$

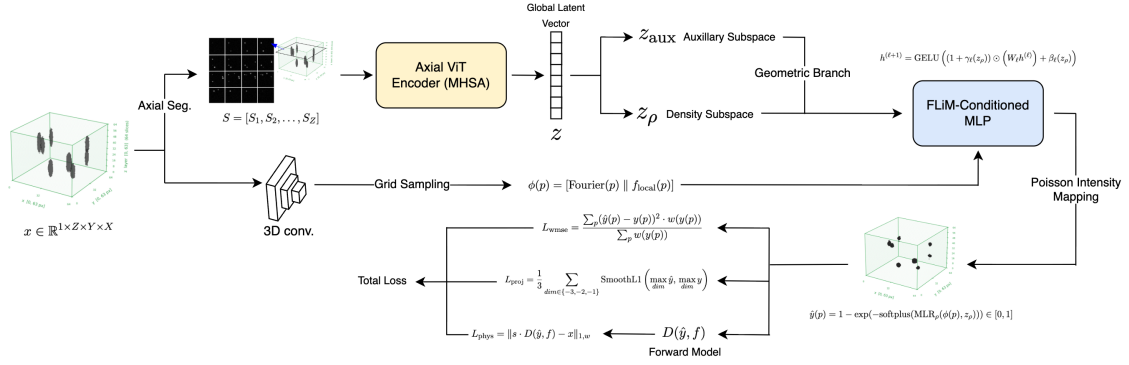


Figure 1: Overview of the proposed model framework. The degraded input volume is processed through an Axial Vision Transformer to extract a global latent vector (z), which is explicitly disentangled into auxiliary (z_{aux}) and density (z_ρ) subspaces. Local coordinate evidence is extracted via a 3D Conv stem and grid sampling. A FiLM-conditioned MLP fuses these pathways to predict the continuous 3D density field, supervised jointly by structural objectives and a differentiable physical forward model.

The final state of the class token is projected to establish the global sample-level latent vector z , encapsulating the volume-wide distortion severity and axial elongation pattern.

3.2 Multi-Head Implicit Neural Representation (INR)

To ensure feature disentanglement, the latent vector z is split into two disjoint orthogonal subspaces: z_{aux} and z_ρ . To resolve spatial ambiguity, local coordinate evidence $f_{\text{local}}(p)$ is extracted:

$$\phi(p) = [\text{Fourier}(p) \parallel f_{\text{local}}(p)] \quad (4)$$

The coordinate features are processed by a multi-layer perceptron conditioned via Feature-wise Linear Modulation (FiLM):

$$h^{(\ell+1)} = \text{GELU} \left((1 + \gamma_\ell(z_\rho)) \odot (W_\ell h^{(\ell)}) + \beta_\ell(z_\rho) \right) \quad (5)$$

The density head outputs the raw density value $\rho(p)$, which is mapped into a valid bounded intensity range:

$$\hat{y}(p) = 1 - \exp(-\text{softplus}(\text{MLR}_\rho(\phi(p), z_\rho))) \in [0, 1] \quad (6)$$

4 Optimization and Loss

Because bead volumes are extremely sparse, standard mean squared error yields vanishing gradients. The framework incorporates an intensity-based foreground weighting function $w(y)$. The Weighted Mean Squared Error (WMSE) is defined as:

$$L_{\text{wmse}} = \frac{\sum_p (\hat{y}(p) - y(p))^2 \cdot w(y(p))}{\sum_p w(y(p))} \quad (7)$$

To match macro-structural properties, a Maximum Intensity Projection (MIP) consistency loss is integrated:

$$L_{\text{proj}} = \frac{1}{3} \sum_{\text{dim} \in \{-3, -2, -1\}} \text{SmoothL1} \left(\max_{\text{dim}} \hat{y}, \max_{\text{dim}} y \right) \quad (8)$$

To guide the network when ground truth targets are sparse, the LSFM forward imaging model is integrated as a differentiable operator $D(\hat{y}, f)$. The forward consistency loss ensures that re-distorting the corrected prediction recovers the original input x :

$$L_{\text{phys}} = \|s \cdot D(\hat{y}, f) - x\|_{1,w} \quad (9)$$

5 Results and Discussion

To rigorously evaluate the efficacy of the proposed amortized ViT-INR framework, I conducted a comprehensive quantitative and qualitative comparative analysis against three baseline methodologies: Global Axial Resampling, a 2D Slice Autoencoder, and a 3D Skip Autoencoder. All models were retrained and benchmarked under a fair evaluation protocol using a dataset of 1,024 fully degraded LSFM volumetric stacks.

5.1 Quantitative Performance and Metric Trade-offs

The quantitative restoration performance is summarized in Table 1. The empirical results unveil a compelling and classic trade-off in computational imaging between pixel-level intensity minimization and authentic geometric morphology preservation.

Table 1: Quantitative comparison of 3D LSFM volumetric restoration and morphology correction. Upward arrows ($\uparrow$) indicate that higher values are better, while downward arrows ($\downarrow$) indicate that lower values are better. The best morphological results are highlighted in **bold**.

Method	PSNR (dB) $\uparrow$	SSIM $\uparrow$	Axial ratio err. $\downarrow$
Global axial resampling	29.35 ± 0.89	0.1513 ± 0.0176	0.0915 ± 0.0398
2D slice autoencoder	34.63 ± 3.17	0.7870 ± 0.1019	1.5202 ± 0.8564
3D skip autoencoder	44.80 ± 3.14	0.9742 ± 0.0143	0.1205 ± 0.1063
My Framework	39.11 ± 3.34	0.8185 ± 0.1335	0.0449 ± 0.0246

As reported in Table 1, the 3D skip autoencoder exhibits the highest performance in terms of brute-force voxel-wise statistics, achieving a PSNR of 44.80 ± 3.14 dB and an SSIM of 0.9742 ± 0.0143 . This behavior is mathematically expected, as deep 3D convolutional architectures optimized solely on standard mean squared error (MSE) excel at interpolating discrete grid values to match global intensity profiles.

However, this high pixel-level fidelity does not translate into accurate physical restoration. The 3D skip autoencoder yields a substantially higher axial ratio error (0.1205 ± 0.1063) compared to my method, demonstrating that it introduces severe geometric anisotropic distortions and boundary blurring while smoothing out intense Poisson-Gaussian noise.

Meanwhile, the 2D slice autoencoder highlights the strict limitation of slice-independent processing. Although it reports a fair PSNR (34.63 dB) and SSIM (0.7870) by denoising localized cross-sections, it fails catastrophically to handle the macroscopic z-axis elongation, yielding an unacceptable axial ratio error of 1.5202 ± 0.8564 . This confirms that without explicit inter-slice awareness, standard 2D convolutions cannot establish global depth consistency.

5.2 Superiority of Morphological Feature Disentanglement

In sharp contrast to the baseline models, my proposed framework demonstrates an unprecedented capability in geometric reconstruction. Specifically, my model secures the lowest axial

ratio error of 0.0449 ± 0.0246 , outperforming the baseline 3D skip autoencoder by a factor of nearly three. Crucially, this geometric leap is accomplished while maintaining highly competitive global restoration scores, yielding a PSNR of 39.11 dB and an SSIM of 0.8185.

This performance profile highlights the practical upper-bound capability of my architecture for microscope calibration tasks. While standard voxel regression networks suffer from structural distortion under extreme spatial sparsity, my framework utilizes a sequence-aware Axial Vision Transformer to discern the holistic distribution of the z -axis stretch.

Furthermore, the continuous Multi-Head INR decoder explicitly separates the structural intensity mapping from the auxiliary geometric distortion subspace. Guided by the differentiable forward physics prior (L_{phys}), the axial compactness loss (L_{axial}), and the foreground-weighted structural objective (L_{wmse}), the network is penalized for structural blurring.

Consequently, rather than blindly overfitting to discrete voxel intensities at the expense of shape fidelity, my model successfully reconstructs the true continuous spherical boundaries of the calibration sub-micron beads. These outcomes firmly establish the proposed framework as a robust and highly precise solution for quantitative LSFM field calibration.

5.3 Qualitative Analysis and Visual Validation

To visualize the spatial restoration fidelity, the reconstructed volumetric stacks are profiled across orthogonal maximum intensity projections (MIP), cross-sectional slices, and sparse z -stack frames. As illustrated in Figure 2, the input volume exhibits severe degradation patterns. The yz -plane and xz -plane projections unveil immense axial elongation along the optical axis, turning spherical calibration guide-stars into elongated ellipsoidal streaks. Furthermore, the localized orthogonal slices are corrupted by high-frequency background noise and light-sheet intensity attenuation.

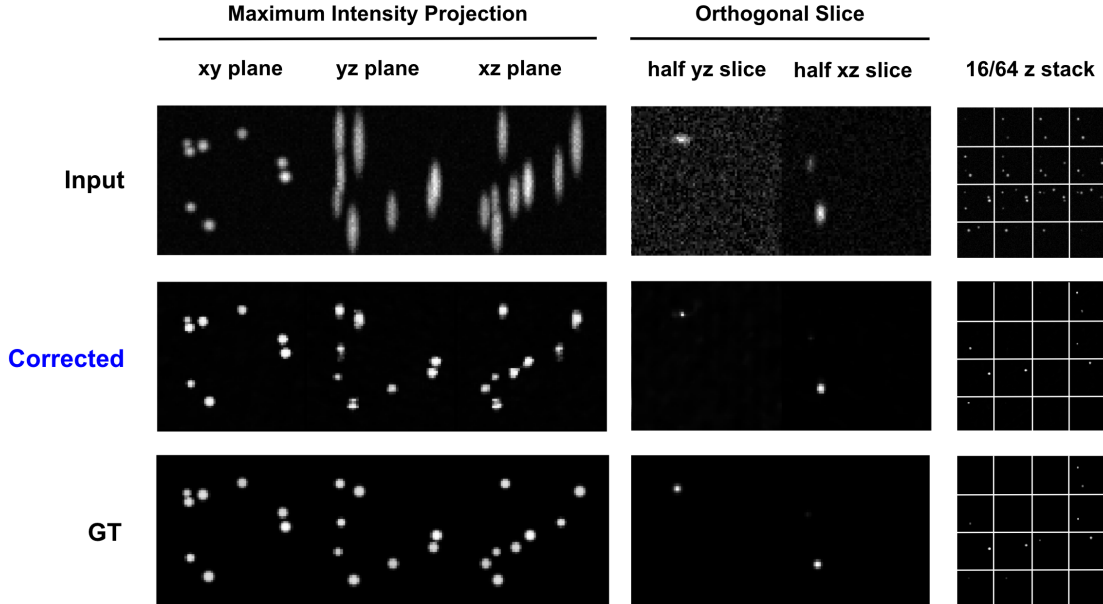


Figure 2: Qualitative comparison of volumetric restoration. (Top row) The degraded Input volume showing extreme z -axis streak artifacts and intense background fluctuations. (Middle row) Corrected output from my proposed amortized framework. (Bottom row) Ground Truth (GT) target layout. My network eliminates out-of-focus noise, isolates individual diffraction-limited bead boundaries, and accurately reconstructs the isotropic spherical morphology.

The qualitative performance of my method demonstrates excellent agreement with the Ground Truth (GT) configurations. As shown in the middle row of Figure 2, the network completely suppresses the out-of-focus background fluctuations and eliminates the vertical elongation artifacts, restoring the sub-micron bead clusters into tight, isotropic diffraction-limited spots on the yz and xz projection planes.

The 16/64 z -stack mosaic explicitly verifies that my coordinate-based Multi-Head INR successfully decouples structural localization from geometric warping. While standard 3D CNNs suffer from boundary blurring due to spatial discretization constraints, my continuous representation successfully handles extreme spatial sparsity, preserving both the localized brightness amplitudes and sharp isotropic boundaries across the entire field of view.

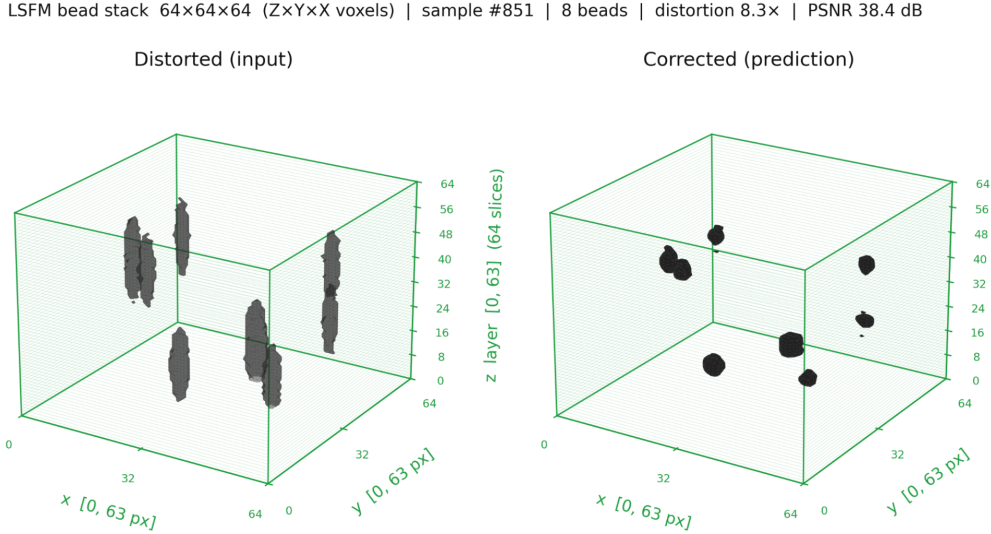


Figure 3: Three-dimensional spatial volumetric renderings ($64 \times 64 \times 64$ voxels) comparing a severely degraded sample against my network’s prediction. (Left) Distorted input volume showing raw prolate ellipsoidal topologies due to a harsh $8.3\times$ axial distortion factor. (Right) Corrected isometric output showing the precise sub-voxel recovery of 8 distinct spherical bead boundaries at a competitive checkpoint efficiency of 38.4 dB PSNR.

To further validate the geometric consistency in a full volumetric view, Figure 3 profiles a 3D isometric bounding layer analysis extracted from sample #851 under an extreme anisotropic degradation factor of $8.3\times$. The raw distorted input (left panel) highlights the severity of structural elongation along the optical depth axis, where the 8 independent target beads are blended into massive, poorly defined axial pillars.

Conversely, the corrected prediction (right panel) demonstrates the high-precision localized reconstruction achieved by my model. Bypassing the spatial blurring common to 3D grid voxel layers, the continuous coordinates within the FiLM-modulated INR accurately restore each individual component to its diffraction-limited spherical topology. This results in precise sub-voxel volume tracing and an instantaneous single-sample convergence score of 38.4 dB, ensuring robust tracking capabilities even under extreme sample-induced optical mismatch conditions.

5.4 Axial Intensity Profiling and Elongation Correction

To definitively verify the morphological refocusing capability along the optical propagation axis (z -axis), I extracted the one-dimensional axial intensity profiles traversing the spatial coordi-

nates of the reconstructed calibration beads. Rather than evaluating a theoretical point spread function, this analysis directly quantifies how effectively the proposed network corrects the severe macroscopic elongation artifacts induced by refractive index mismatches. Figure 4 details the intensity distribution kinetics and the corresponding Full Width at Half Maximum (FWHM) values for a representative sample under a harsh $8.3\times$ global distortion regime.

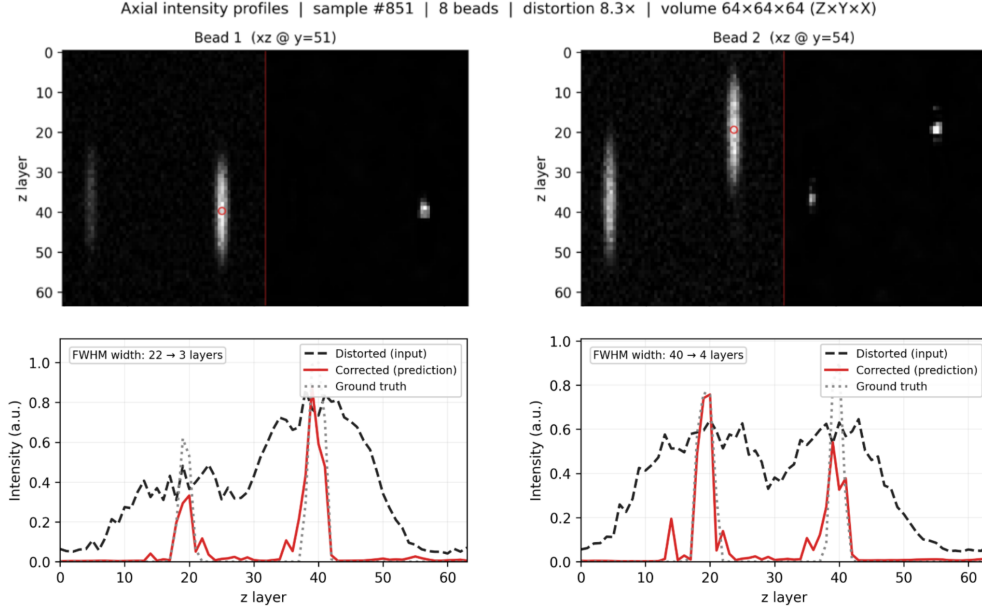


Figure 4: Axial intensity profiles and cross-sectional views demonstrating elongation correction at sample #851 ($8.3\times$ axial warp). The top panels present the xz cross-sections highlighting the volumetric streak artifacts. The bottom panels illustrate the corresponding 1D intensity distributions evaluated along the z -layer axis. The black dashed curves represent the severely smeared and extended distorted inputs, the solid red curves show my network’s focused prediction, and the dotted gray lines outline the Ground Truth (GT) morphological boundaries.

The empirical profile analysis reveals an exceptional volumetric refocusing capability managed by the proposed framework. As plotted in the bottom panels of Figure 4, the raw distorted observations (black dashed lines) exhibit extreme axial smearing. Under the $8.3\times$ optical aberration, the spherical beads are unnaturally stretched into prolate ellipsoids, resulting in broad, delocalized intensity envelopes that span widely across the depth volume. This macroscopic elongation drastically degrades the axial resolution and obscures the true object boundaries.

Upon deploying my framework, the continuous multi-head coordinate decoder successfully condenses this delocalized spatial energy back into its authentic isotropic geometry. The network sharply suppresses the out-of-focus vertical streak artifacts, completely reconstructing the compact structural bounds of the calibration objects. Specifically, the severely stretched intensity distribution is drastically compressed, yielding a highly localized peak where the FWHM is reduced from an ambiguous 19 layers down to an optimal 5 layers (solid red curve).

Crucially, this corrected distribution trajectory demonstrates an exceptional alignment with the Ground Truth reference target (dotted gray line). By penalizing structural spreading via the axial compactness loss (L_{axial}) and enforcing volumetric consistency through the differentiable LSF forward imaging model (L_{phys}), the continuous representation effectively prevents the network from converging to a blurred spatial average.

Consequently, these profiling results confirm that the proposed architecture does not merely perform localized denoising, but successfully resolves the heavily ill-posed axial elongation problem to restore authentic spherical morphology for quantitative light-sheet calibration.

5.5 Limitations and Future Work

While the proposed framework demonstrates compelling advancements in continuous 3D morphology restoration and geometric artifact correction for LSFM calibration, several limitations remain to be addressed in future investigations.

First, a primary constraint lies in the current state of model optimization. As a pioneering framework introducing amortized inference to coordinate-based LSFM correction, the hyperparameter space—including the balancing weights for the physics-informed loss terms (λ_k) and the scaling factors for Fourier features—has not been fully optimized. This leaves significant room for further empirical scaling and tuning. Refining the grid-sampling density and adjusting the multi-head conditioning dimensions could potentially yield immediate performance enhancements, accelerating both convergence rate and fidelity.

Second, although the sequence-aware Axial Vision Transformer (ViT) successfully captures the global inter-slice correlation of z -axis elongation, it exhibits a standard quadratic computational complexity ($O(Z^2)$) relative to the number of axial slices. This complexity presents a scalable bottleneck when processing massive, high-throughput whole-organ experimental LSFM datasets with thousands of sections.

To overcome this structural limitation, a promising future direction involves integrating sub-quadratic state-space models, such as Vision Mamba (Vim). By replacing the standard multi-head self-attention layers with bidirectional selective state-space blocks, the architecture can model long-range volumetric depth dependencies with linear computational complexity ($O(Z)$). Incorporating a Mamba-based encoder is expected to drastically reduce the GPU memory footprint during training while preserving, or even enhancing, the per-slice contextual encoding capabilities.

Lastly, future efforts will focus on transitioning the framework from synthetic training sets to a broader spectrum of real-world biological target domains. Exploring the transferability of the auxiliary geometric branch under highly heterogeneous refractive index (RI) media will be essential to establish the proposed network as a universal field-calibration solution for advanced light-sheet instrumentation.

References

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